

Comparing Minimum Wage to the Cost of Living

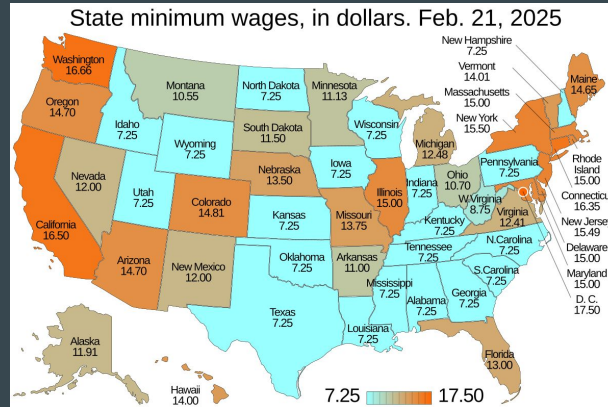


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Project Goal

Our project explores the gap between minimum wage and the cost of living across different U.S. regions. Using data from the Bureau of Labor Statistics (BLS) and the Consumer Price Index (CPI), we analyze whether workers earning minimum wage can afford essential expenses like housing, food, and transportation. Our goal is to visualize wage trends and highlight economic disparities.



The Real-World Impact

- Millions of workers earn minimum wage, but can they afford basic necessities?
- Cost of living varies, but wages don't always adjust.
- Inflation reduces purchasing power, pushing many into financial hardship.



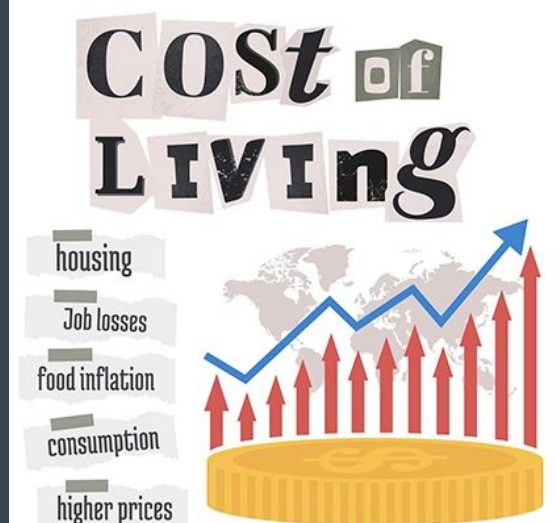
Social & Economic Importance

- Many workers struggle to afford housing, food, and healthcare.
- Governments use this data to propose minimum wage adjustments.
- Higher wages can boost consumer spending and economic growth.



Key Questions

- Can a full-time minimum wage worker afford basic necessities?
- How does inflation impact the real value of minimum wage over time?
- Which states have the biggest gap between wages and cost of living?
- What policies could help close this gap?



Approach

Technology Used: We designed an interactive notebook in [Observable](#), using Vega-Lite for data visualization.

Intended Audience:

- Policymakers analyzing wage fairness & inflation impact.
- Economists and researchers studying cost-of-living trends.
- General public interested in how inflation affects their earnings.

Data Sources:

U.S. Bureau of Labor Statistics (BLS) Minimum Wage Data

- Tracks minimum wage across states.
- Widely used for labor policy analysis.

Source: [BLS Minimum Wage Report](#)

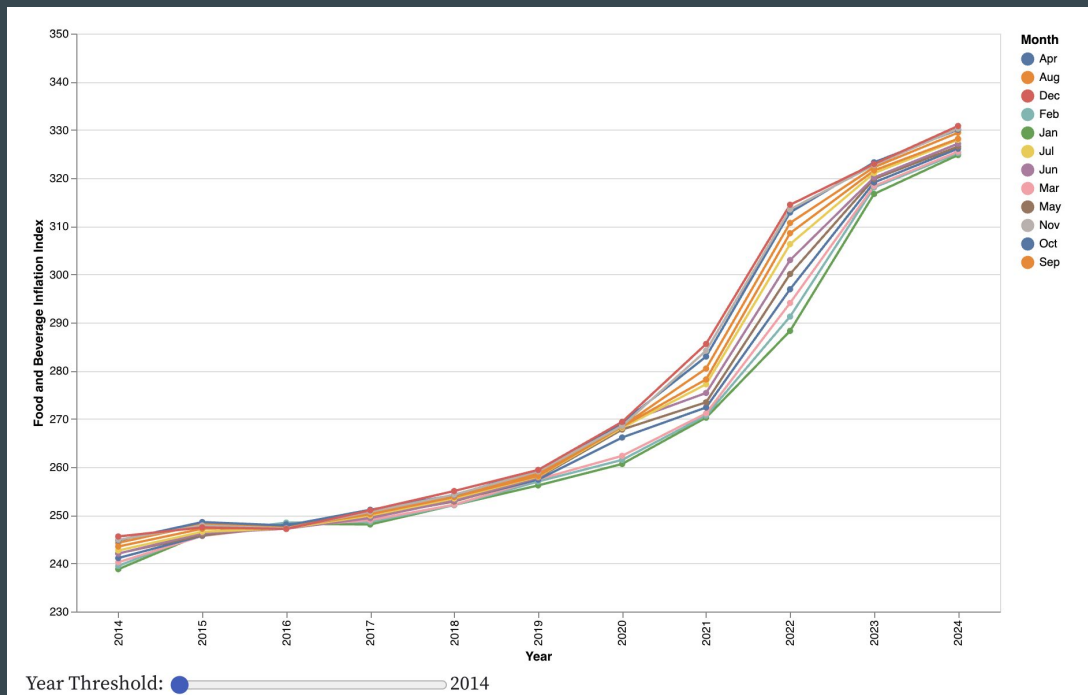
U.S. Bureau of Labor Statistics Consumer Price Index (CPI)

- Measures inflation trends.
- Used to analyze purchasing power over time.

Source: [BLS Consumer Price Index](#)

Interactive Wage Trend Visualization

- Average effective minimum wage trends in the U.S. (2014–2024).
- Slider: Adjust the time range to explore specific years.
- Trend Analysis: Observe how wages change over time.



Conclusion

Our interactive analysis provides insights into minimum wage trends, but it does not fully account for regional differences in essential living costs such as housing, healthcare, and transportation. While we incorporate inflation data through the Consumer Price Index, further adjustments are needed to assess real purchasing power over time. Future work could expand the dataset to include cost-of-living indices, add interactive features for state-level analysis, and implement predictive modeling to forecast wage trends. These enhancements would provide deeper insights for policymakers, researchers, and individuals analyzing the economic challenges faced by minimum wage workers.

Summary

In summary, we built an interactive notebook in Observable to analyze minimum wage trends from 2014 to 2024, using Vega-Lite visualizations. Our work allows users to explore how wages have changed over time and highlights the gap between minimum wage earnings and the rising cost of living. By combining data from the U.S. Bureau of Labor Statistics and the Consumer Price Index, our project provides a foundation for further research into wage policy, inflation, and economic disparities. Moving forward, this tool can be expanded to include additional economic indicators, providing a more comprehensive picture of the challenges faced by minimum wage workers in the U.S.